

Traffix.

Intelligent Traffic Congestion Response

A shared view of traffic. A safer way to test the response.

 LIG Square, Indore



Traffic changes. Fixed responses fall behind.



Mixed vehicles move at different speeds.



A blocked exit can spread a queue.



Sparse observations hide early congestion.



An untested diversion can move the problem.



Operators need evidence and a way to compare responses.

One simulation, connected decisions

Observe

Source-labelled sensors
and opted-in probes



Detect

Queues, speeds and
fresh evidence



Respond

Bounded signals and
driver-accepted routes



Compare

Matched runs against
fixed timing



What makes Traffix different



Mixed Indian vehicle classes on LIG geometry



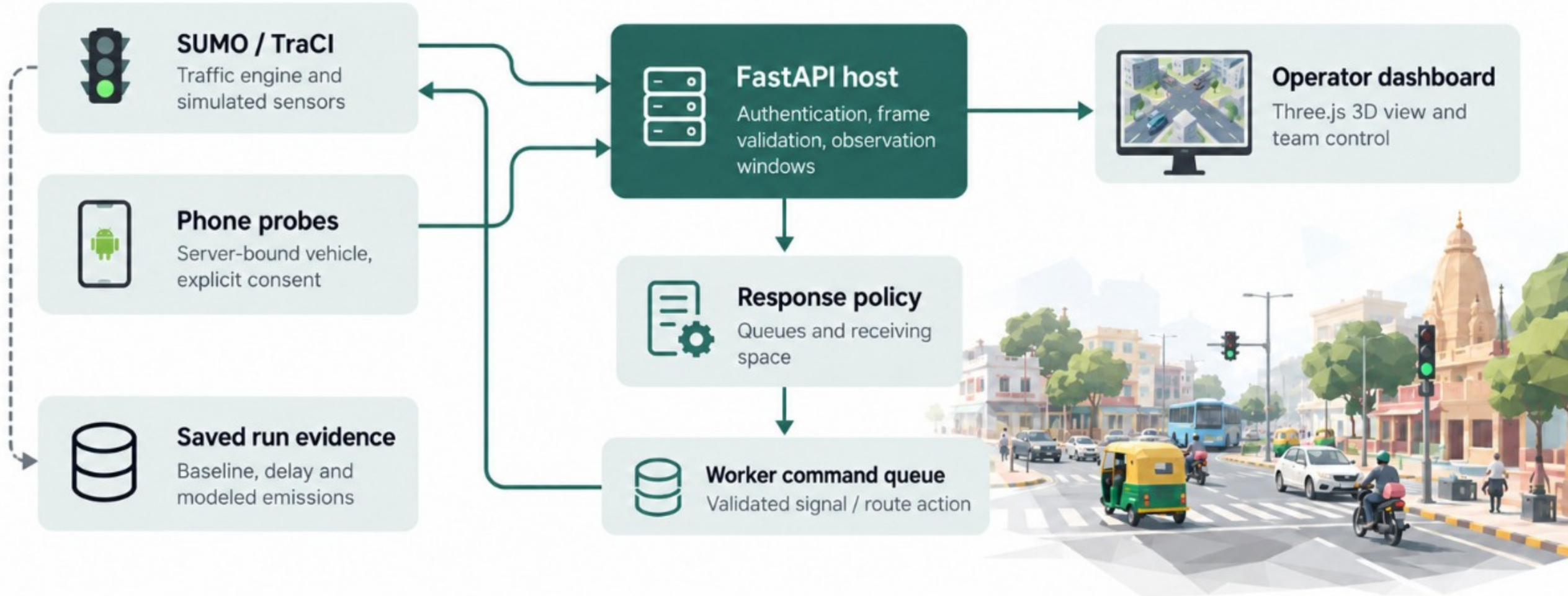
One vehicle identity across phone and dashboard



Unknown stays unknown. Results stay auditable.



One host owns the simulation



Capacity-aware pressure heuristic

Serve demand without feeding a blocked exit.

- 1 Read fresh movement queues and downstream occupancy
- 2 Score upstream queue pressure against downstream load
- 3 Choose a legal phase with receiving space
- 4 Apply through yellow and all-red clearance



Minimum / maximum green



Maximum-red fairness



Hysteresis reduces switching



Stale data triggers fallback

Detection and forecasting

Implemented detection

Fresh slowdown persists for 30 seconds.
Evidence must continue through a serving green.
Sparse or stale observations remain unknown.

Recovery needs 60 seconds of low slowdown.
Defaults are assumptions requiring calibration.

Forecasting validation plan

HistGradientBoostingRegressor
120, 180 and 300 second horizons
Causal observation history, age and coverage

Hold out whole runs and seeds. Compare MAE
and usable coverage with persistence and trend.

Measured signal savings come from reactive pressure control. Predictive-control benefit remains unvalidated.

Held-out simulation results

11.31%

shorter mean journey

7.48%

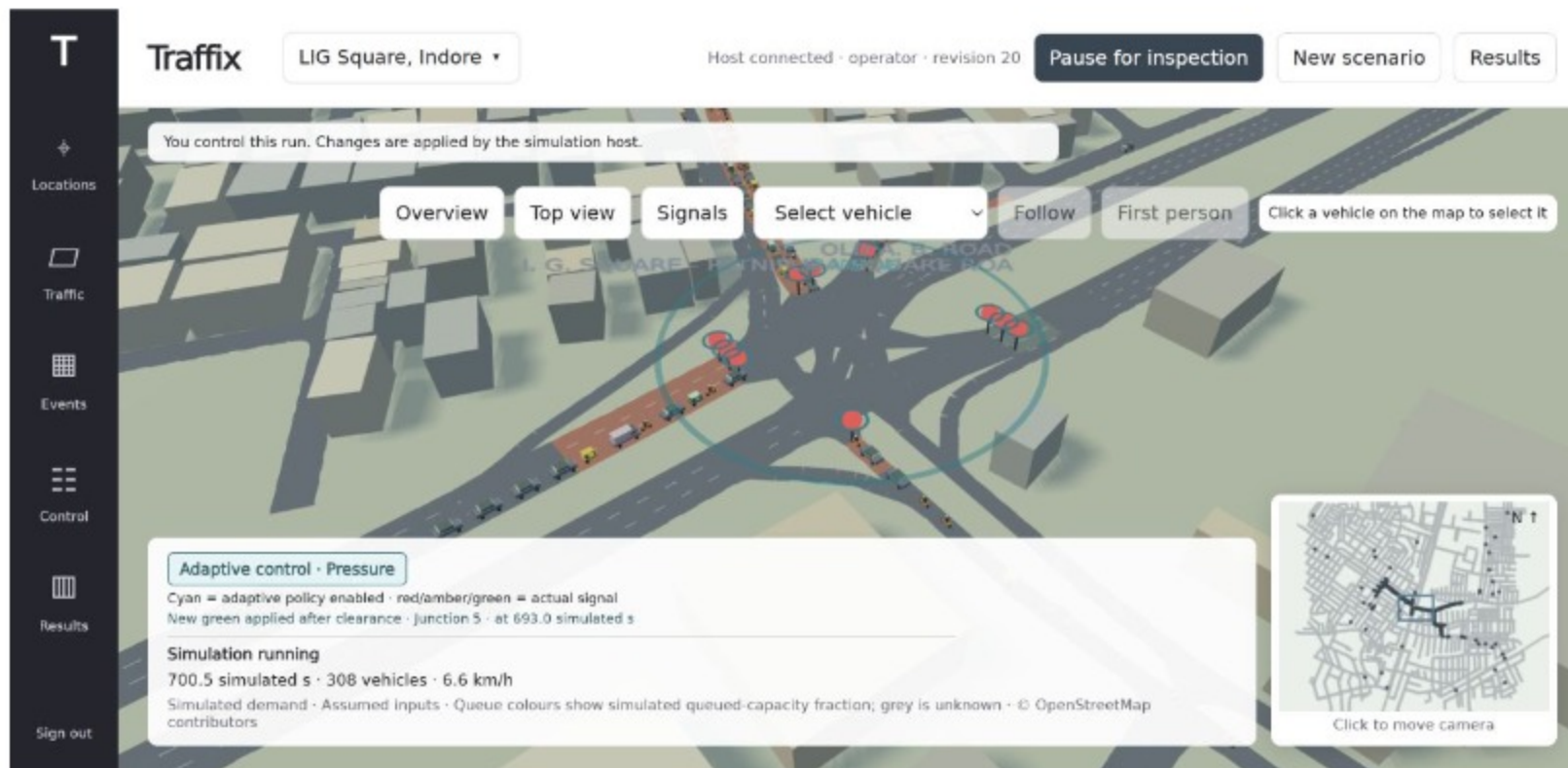
lower modeled CO₂

Held-out seed	Fixed journey	Pressure journey	Journey reduction	CO ₂ reduction
51 (49 vehicles)	291.83 s	259.67 s	11.02%	6.76%
52 (79 vehicles)	329.67 s	291.43 s	11.60%	8.21%

128/128 scheduled trips completed under each policy. Zero collisions and teleports.

Mean of paired seed reductions versus fixed timing. Synthetic finite cohorts, not field effectiveness.
SUMO modeled emissions. Two seeds only. Incomplete stress baseline excluded from percentage savings.

Operator and driver demonstration



One shared host

Admin selects a policy.
Worker confirms safe actions.
Results compares saved cohorts.

Driver joins a bound vehicle.
Consent controls probe sharing.
Guidance requires acceptance.

Saved UI screenshot. Phone observations relay simulated vehicles. Native integration and physical multi-device checks require verification.

Validation and deployment roadmap

Built and measured

Local LIG simulation, shared operator dashboard and bounded pressure control.
Saved matched cohorts support the held-out result.

Next integration gate

Verify the native companion against the unified host on real devices.
Keep phone identity, consent and worker confirmation visible in the demo.

Next evidence gate

Validate forecasts, sustained overload and per-approach fairness.
Calibrate fleet emissions and observations before a field pilot.